Design review · status after V8.2

Taif / Palm House · the review of 30 Oct 2026 (`docs/review/design_review.typ` → `output/Design_Review_V6.pdf`, tag `review-1`, written against V6.2) · status on 3 Nov 2026 after V6.3, V7.0, V7.1, V8.0 (fire safety, courtyard layout, air system, south wing v2), V8.1 (the roof ladder, the lowered roofline, the office from the suite court, one site entry) and V8.2 (a checkpoint: the stair flip, bedroom 1 without a bath, the court's mashrabiya, the clerestories openable, the food route accepted, the site and services plan, the plumbing study rounds 1–2 as proposals); the rows updated in V8.2: X-01 closed (accepted), S-04 not needed, E-05 no generator, S-01, S-02, S-05, S-08, E-03, E-04, E-06, G-02, G-07 · every critical and major finding: 30 in all, **resolved** 20 · **partly** 8 · **open** 2

Id	Sev.	Where	Status	Version	What was done · what is still open
Y-01	critical	Courtyard · the envelope	**resolved**	V7.0 / V7.1	Enclosure option A: W1 on the gallery edge, W2 on the family living's edge, W3 over the living's courtyard glass, black steel divided lites, the projection 1.55 m (B3 / R11 / C21–C23 at x 5.80); the courtyard open to the sky. V7.1: semicircular heads over the three-arch arcade.
A-01	critical	Ageing · the lift	**resolved**	V6.3	The lift installed at construction, two stops; the shaft a placeholder until a supplier's drawing (open item 15: headroom ≤ 3.40 m / ≤ 3.70 m with a raised head).
S-03	critical	Site · north and lane setbacks	**partly**	V6.3	The house 0.30 m south: 0.20 m to the lane setback line, 0.80 m to the north street line, provisional. **Open:** the legal survey (open item 1).
E-08	major	Services · fire and life safety	**resolved**	V8.0 / V8.1	Home-level strategy (owner): interconnected mains smoke alarms with battery back-up in every bedroom and on the routes, heat, CO and gas detectors, extinguishers and a blanket; escape casements without screens in every sleeping room (the maid room's at sill +1.10); thumb-turns on the exits (`studies/fire_escape.typ`). V8.1: the first floor's exits are the main stair plus the escape window of every sleeping room; the suite court is a fresh-air area, not an exit; the roof ladder is no escape device.
G-01	major	GF · living wall (west)	**resolved**	V6.3	The solid TV wall with the entrance arch and the arched TV niche.
G-04	major	GF · maid room	**resolved**	V7.0	The kitchen's north wall at y 7.0: maid room 7.28 m² clear, 2.49 m at the bed.
F-01	major	FF · gallery width	**resolved**	V7.0	The projection 1.55 m: the gallery 1.405 m clear, 1.235 m at the column sleeves.
F-02	major	FF · door widths	**resolved**	V6.3	Bedroom and bathroom doors 1.00 m openings; exceptions recorded.
F-06	major	FF · family living	**resolved**	V7.0	W2 on the courtyard edge: a conditioned room.
F-07	major	FF / roof · roof access	**resolved**	V7.0 / V8.1	V8.1: the roof by a manufactured maintenance ladder in a 0.90 × 0.30 recess of a stone pilaster on the gym's east wall in the suite court (3.70 m, the stiles 1.10 m above the roof, a self-closing gate in the court guard, a fall-arrest rail), behind a locked walnut panel (`studies/roof.typ` p. 2); the guarding continuous to every piece of equipment; the internal stair stops at the first floor, the roof slab closed over the well. **Open:** the fall protection confirmed by the specialist (open item 4).
X-01	major	Circulation · serving the majlis	**resolved**	V8.1 (owner)	**Accepted by the owner** (2 Nov 2026): the food route from the kitchen through the dining and the riwaq to the serving passage stays as drawn; normal guest numbers, very large events use the old majlis on the farm, the owner serves. No outside service path, no serving door in the majlis (open item 19 closes).
G-02	major	GF · living + dining	**partly**	V7.0 / V8.2	The courtyard well decided; V8.0: the clerestories get the winter noon sun on their whole height; V8.2: the clerestories stay openable (decided), the garden door not needed (S-04, owner). **Open:** the daylight simulation (open item 15).
G-03	major	GF · kitchen	**partly**	V8.0	The hood's make-up air: a filtered motorised inlet under the stone band on the east wall, interlocked with the hood (air v3.1). **Open:** the MEP sizing (open item 9).
G-07	major	GF · bathrooms	**partly**	V8.2	The first-floor showers reachable by the lift; the ground-floor shower studied (`studies/gf_shower.typ`: P1 the family WC + basins as one accessible wet room, preferred) and **deferred to the plumbing study** (owner, 2 Nov 2026), where P1 is the baseline of scenario B. **Open:** the owner's decision with the plumbing scenario (open item 10).
Y-02	major	Courtyard · size for daily use	**resolved**	V8.0	The courtyard layout (owner): the seating one step down with three benches for six to eight and a low table; the TV wall; children's play in the garden, not in the courtyard; a hose point at the enclosure (`studies/courtyard_layout.typ`).

Y-03	major	Courtyard · sunken seating safety	**resolved**	V8.0	The 0.45 m pit replaced by one 0.15 m step with a honey-stone nosing and a riser light; level strips ≥ 0.6 m at every glass line; the drain and an independent overflow (the drainage section).
T-01	major	Structure · X5	**partly**	V8.0	Load path revised, **improved, not verified**: R6 removed (the suite court open to the south facade); the gym's east wall on B19 between C19 and C20, the double wall on B8 and a new B22, the office wall non-load-bearing on rib B21. **Open:** the engineer confirms (open item 3).
T-03	major	Structure · soil and foundations	**open**		No soil report (open item 7): the pads, the lift pit and the breeze channel are assumed.
E-01	major	Services · the malqaf's value	**resolved**	V8.0	Air system v3 / v3.1 (owner): one tower with two separate paths: a gentle courtyard breeze (a damper, a shallow channel in the slab, the grille under the TV, an assist fan) and the fresh-air intake of the HRV in the tower's base on the roof; no water, no pit, no entry (`studies/air_system.typ`). The MEP engineer sizes it (open item 9).
E-02	major	Services · ventilation and dust	**partly**	V8.0	A 750 m^3/h class HRV with an ISO ePM1 ≥ 50 % filter, continuous + boost, a dust mode, the kitchen make-up air (all assumed). **Open:** the MEP design.
E-03	major	Services · plumbing	**partly**	V8.2	The plumbing study rounds 1–2 done (`studies/plumbing.typ`, proposals): two clusters, buried GF runs north to the septic near C1 with the inverts computed, the suite bath over the kitchen in sections with (b) an east service wall + the tub's plinth recommended, the east stack external behind cladding, rain and sewage fully separate, the condensate connections, the smells list, the utility room layout. **Open:** the owner's decisions (the scenario, (b), P1, the stack, the heaters) and the plumbing engineer's design (open items 10, 11).
E-04	major	Services · solar PV provision	**resolved**	V8.1 / V8.2	PV zones inside the guarded zones (9.8 m^2 south wing, 20.5 m^2 north block); V8.2: the conduits down the west shaft to the utility room, where the inverter and the battery cabinet go (owner; the electrical engineer confirms space, ventilation and fire safety; the layout with the water kept away in the plumbing study). More PV area if wanted later.
E-05	major	Services · generator and essentials circuit	**resolved**	V8.2 (owner)	**No generator** (owner, site plan): the backup comes from the PV and the batteries when installed; until then no backup during power cuts (the gas valve fails closed); an essentials circuit at the main panel for the later batteries (R-32 changed).
E-06	major	Services · security and the safe room	**resolved**	V8.1 / V8.2	No safe room (owner, R-33); one site entry, the farm's 4 m main gate; office visitors met in the men's majlis; the roof ladder behind a locked panel. V8.2 (site plan): a video intercom and camera C1 at the gate, cameras C2–C7 round the house, the recorder and the alarm panel in the utility room, keypads at both entrances; positions proposed, the electrical engineer designs it.
E-07	major	Services · acoustics of the living	**open**		The double-height living's finishes and absorption at the detail round.
S-01	major	Site · parking	**resolved**	V8.2 (owner)	The site plan: the family's existing covered parking for two cars south of the old house (beyond the plot) and one uncovered family car in the north setback strip, kept (the municipality allows it; a 3-point manoeuvre from the driveway, the entrance clear); guests in the farm parking. R-11 partly (the walk from the covered parking uncovered).
S-02	major	Site · guest approach	**resolved**	V8.2	The guest path: a 1.2 m honey-stone band along the driveway's east edge from the farm parking to the guest entrance (15.5 m, bollard lights), never entering the family areas; the garden behind its 1.2 m wall.
S-04	major	Site · garden access	**resolved**	V8.2 (owner: not needed)	No garden door from the living / dining: the garden (south only) is reached from the kitchen's service door round the south-east corner or from the guest entrance side.
S-05	major	Site · east strip, service yard	**resolved**	V8.2	The service yard at the kitchen's service door: the bins, the 1.8 m screen, the foul chamber at its south end, the LPG cabinet on the screen's outer face (≥ 1.5 m from every opening, ≥ 3 m from the manhole, the rule stated); the workers' path from the gate along the north and lane-side strips; the hedge closes the yard from the garden; the outdoor counter and the grill as V8.1.
S-08	major	Site · tanks, septic, plant	**partly**	V8.2	The site plan (owner's information): the existing water tank just east of peg R with the reserved future tank beside it; the existing septic outside the farm wall near corner C1, the foul drains rerouted north and west to it with the inverts computed (a pump station if the surveyed inlet is higher); the roof plant as V8.1; no generator. **Open:** the survey of the septic (position, inlet level,

					capacity, condition), the tanks and the levels (open item 16); the municipality's consent (17).

The next open items, in this order

1. **The owner's plumbing decisions** (E-03, G-07; open items 10, 11): the suite bath over the kitchen as (b) a service wall + the tub's plinth, the ground-floor shower P1, the east stack external behind removable cladding, the heaters and the batteries with the engineers, the utility room's door.
2. **The survey of the septic, the tanks and the levels** (S-08; open items 16, 1): the septic outside the wall near C1 (its inlet level decides a pump station), the tank east of peg R, peg R and C1, the street level at the gate.
3. **The engineers:** structure X5 and the roof ladder (T-01, F-07; open items 3, 4), the soil report (T-03; 7), MEP sizing of the air system, the HRV and the plumbing (E-01 … E-03; open items 9, 10).
4. **The legal survey** (S-03, open item 1) and the municipality (open item 17: the septic outside the wall, the soakaways, the gas cabinet, the Civil Defense).
5. **V9.0** with the full verification and validation review.
6. **The living's acoustics and the facades** at the detail stage (E-07).

The full verification and validation review is planned for V9.0, after the owner's plumbing decisions and the survey. This page is a status update, not a new review.

Minor findings, suggestions and strengths are unchanged in the review; of the minors, T-02 (X3) is resolved (V6.3), F-09 (the gym's treadmill clearance) is resolved in V8.0 (the manufacturer-type zone on the west band, vibration with the engineer), Y-04 (courtyard dust and water) is answered by the drainage section and the hose point, F-05 (bedroom 1's bath) is resolved in V8.2 (option A, owner: no private bath, night lights on the route). Nothing here is for construction: concept design, all sizes assumed until the engineers confirm.